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clear
clc
close

% Load simpleGameEngine and Sprite Sheets
sprite_sheet = 'spritesheet.png';
sprite_sheet2='spritesheetC.png';

sprites = simpleGameEngine(sprite_sheet, 33, 33,3,[0,0,255]); % Load 33x33 sprites
of game pieces, 3 zoom, and matching background color
menuSprites=simpleGameEngine(sprite_sheet2, 33, 33,5); % load 33x33 sprites of menu
pieces, 5 zoom

% variables for sprites
Empty = 1;
Red = 2;
Yellow = 3;

% make the Connect 4 board (6 x 7)
rows = 6;
collumns = 7;
board = ones(rows, collumns) ; % start with board empty

% player turn, 1 for Red, -1 for Yellow
playerTurn = 1;

% draw scene of start menu
drawScene(menuSprites, [1,12,13,14,1;1,1,11,1,1])

% loop to get mouse input
while true
    [r,c] = getMouseInput(menuSprites);
    if [r,c]==[2,3] % if player clicks on start, loop ends
        close
        break
    end
end

% game Loop
while true

    % show current board
    drawScene(sprites, board);

    % displays which player should go
    if playerTurn == 1

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        title('Player 1s turn (Red)')
    else
        title('Player 2s turn (Yellow)')
    end

    % player chooses a column
    collumn = input('Choose a column (1-7): ');

    % check if collumn is available
    if collumn < 1 || collumn > collumns || board(1, collumn) ~= Empty
        disp('Please choose a column between 1 and 7 that is not full. ');
        continue; % returns to the choose collumn input
    end

    % place the disc in lowest row
    for row = rows:-1:1 % checks rows from bottom to top
        if board(row, collumn) == Empty % checks if chosen position is empty
            if playerTurn == 1 % if it is player ones turn, it places a red disc
                board(row, collumn) = Red;
            else
                board(row, collumn) = Yellow; % if it is player twos turn, it places a
yellow disc
            end
            break; % exits current for loop and continues
        end
    end

    % Check for a win condition
    if playerTurn == 1 % checks which player is active and assigns color to
playerDisc
        playerDisc = Red;
    else
        playerDisc = Yellow;
    end
    % runs checkWin function to check for winning conditions
    if checkWin(board, playerDisc)
        drawScene(sprites, board);
        if playerTurn == 1
            close % close connect 4 board
            drawScene(menuSprites, [1,1,9,1,1;1,15,1,16,1]) % draw menu scene
            title('Player 1 (Red) wins!') % title displaying winner

            % loop to get mouse input on menu, to choose to start new
            % game or exit
            while true
                [r,c] = getMouseInput(menuSprites);
                if [r,c]==[2,2]; % if mouse input = 2,2, player wants to start new
game
                    board = ones(rows, collumns); % board resets
                    close %closes figure

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        clc % clears text
    elseif [r,c]==[2,4]; % if mouse input = 2,4, player wants to exit
        close % close figures
        clear % clear workspace
        clc % clear text
    end % end if statement
    break; % breaks out of while loop

end
continue % continues to next loop, restarting game
else
    close % closes figure
    drawScene(menuSprites, [1,1,10,1,1;1,15,1,16,1]) % draw menu scene
    title('Player 2 (Yellow) wins!') % title showing winning player
% loop to get mouse input on menu, to choose to start new
% game or exit
    while true
        [r,c] = getMouseInput(menuSprites);
        if [r,c]==[2,2]; % if mouse input = 2,2, player wants to start new
game
            board = ones(rows, collumns); % board resets
            playerTurn = -playerTurn; % swaps player turn to red to reset
game
            close % close figures
            clc % clear text
            elseif [r,c]==[2,4] % if mouse input = 2,4, player wants to exit
                close % close figures
                clear % clear workspace
                clc % clear text
            end
            break; % break out of while loop
        end
        continue % continues to next loop restarting game
    end
end
% switch player
playerTurn = -playerTurn; % switches player to other color, changes 1 to negative 1
and vice versa
end

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Unrecognized function or variable 'sprites'.

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% function to check for win condition

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function win = checkWin(board, player)
    % check horizontal, vertical, and diagonal lines for 4 consecutive discs
    rows = size(board, 1); % checks size of rows
    collumns = size(board, 2); % checks size of collumns
    win = false; % create win variable and assigns it to false

    % Check horizontal
    for r = 1:rows % runs for loop based on ammount of rows
        for c = 1:collumns-3 %runs for loop based on collumns minus 3, or four in a
row
            if all(board(r, c:c+3) == player) %checks to see if 4 in a row are all
the same player
                win = true; % if 4 in a row are the same, assigns win to true
                return; % exit function and loop if win = true
            end
        end
    end

    % Check vertical
    for c = 1:collumns % runs loop based on ammount of collumns
        for r = 1:rows-3 % runs for loop based on rows minus 3, or four in a row
            if all(board(r:r+3, c) == player) % checks to see if 4 in a row are
all the same player
                win = true; % if 4 in a row are the same, assigns win to true
                return; % exit function and loop if win = true
            end
        end
    end

    % check diagonal left to right
    for r = 1:rows-3 % runs for loop based on rows minus 3, or four in a row
        for c = 1:collumns-3 % runs for loop based on collumns minus 3, or four in
a row
            if all(diag(board(r:r+3, c:c+3)) == player) % checks to see if four in
a row are the same player diagonally
                win = true; % if 4 in a row are the same, assigns win to true
                return; % exit function and loop if win = true
            end
        end
    end

    % check diagonal right to left
    for r = 4:rows % runs based on last 4 rows
        for c = 1:collumns-3 % runs for loop based on collumns minus 3
            if all(diag(flipud(board(r-3:r, c:c+3)))) == player) % checks to see if
four in a row are the same player diagonally but flips the board horizontally to
check the opposite way from previous check
                win = true; % if 4 in a row are the same, assigns win to true
                return; % exit function and loop if win = true
            end
        end
    end
end

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end  
end  
end
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